

Andrew Winter

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Curriculum Vitae – August 16, 2026

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Overview

I study how the environment surrounding forming stars shapes the protoplanetary discs and planets that emerge from them, connecting star-forming regions, disc evolution, stellar clusters and galactic-scale star formation to observed exoplanet demographics. My work is primarily theoretical, developed in close collaboration with observers and grounded in observational statistics, and has been covered in outlets including [New Scientist](#) and [Forbes](#). I bring broad computational experience, primarily in Python, spanning hydrodynamics (GANDALF, PLUTO), N-body dynamics (NBODY6, Rebound), radiative transfer (RadMC3D) and observational data analysis (Gaia, ALMA/JWST spectral-line fitting).

Research Positions & Funding Awards

- 05/25– **Royal Society University Research Fellowship:** ‘*Environmental origins of exoplanet diversity*’ – Queen Mary University of London, £1.83M
- 09/24–04/25 Visiting researcher – Max Planck Institute for Astronomy, Heidelberg
- 10/23–04/25 **Marie Skłodowska Curie Fellowship:** ‘*Observational signatures of planet formation in irradiated discs*’ – Observatoire de la Côte d’Azur, €195k
- 10/22–10/23 PDRA – Kinematic signatures of planet formation in protoplanetary discs, Observatoire de la Côte d’Azur
- 6/22–7/22 Visiting researcher – Chemical inhomogeneities in globular clusters, Institute of Astronomy, University of Cambridge
- 4/22 **Paris Regional Fellowship** (declined), €130k
- 3/20–3/22 **Humboldt Fellowship** – ‘*Planet evolution in dense stellar clusters*’, Heidelberg University, €80k
- 5/19–3/20 PDRA – ‘*Photoevaporative disc winds around binary stars*’, University of Leicester
- 9/18–11/18 DFG funded visiting Ph.D. student – ‘*Linking galactic-scale star formation physics to protoplanetary disc survival*’, Heidelberg University

Education

- Sep 2026 Certificate in Learning and Teaching (CILT), Queen Mary University of London
- 2015–2019 Ph.D., Institute of Astronomy, University of Cambridge
Graduation date: Oct 2019, Supervisor: Cathie Clarke
Thesis: The influence of stellar birth environment on protoplanetary disc evolution

2011–2015 MMathPhys (Master of Mathematics and Physics), University of Warwick
1st Class Honours, Supervisor: Andrew Levan
Dissertation: Host galaxies of core collapse supernovae

Publications

I have an h-index of 26 and a total of 1847 citations, with 18 first author articles. A full publication list is available on [NASA/ADS](#).

Conferences & Talks

I have given 19 talks in departments across Europe, the US and China and at 42 international conferences since 2019. This includes 9 invited review talks at large international conferences.

Supervision & Teaching

Supervision **PhD:** Oliver Brown (2025–)*, Mika Kontiainen (2024–), Rossella Anania (2022–25), Daniele Fasano (2022–25)
Masters: Fahham Kurji (2026–)*, James Wirth (2023–24)*
Bachelors: Linling Shuai (2022–23)*
Summer: Audrey Liu (2026)*
*Main supervisor (others: co-supervision or informal)

Teaching Physical Cosmology Lecturer/Module Co-organiser, QMUL (2025–present);
Part II Supervisor – General Relativity & Astrophysical Fluid Dynamics,
University of Cambridge (2015–19)

Community & Collaborations

I am an active member of several large collaborations: [exoALMA](#) (ALMA large programme), [Transition Discs](#) (DFG research unit), [EWOCS](#) and [XUE](#). I contribute to the community as a JWST Cycle 4 discussion panelist (previously Cycle 2/3 external panel) and an ALMA distributed peer reviewer, and as a grant reviewer for Fondazione Cariplo, the Humboldt Foundation and STFC, alongside regular refereeing for MNRAS, A&A, AJ, ApJ and Nature Astronomy. I have organised several conferences and workshops, including the PLATO Theory group meeting (QMUL, 2025) and the exoALMA workshop (Nice, 2024).

Observing Time

NOEMA Principal Investigator, 3mm continuum/line survey – 50.6h science time + overheads, Grade B (observations in progress)

ALMA Co-Investigator on 9 accepted proposals (Cycles 2023–2026, Grades B/C) and Principal Investigator on 1 further programme, studying protoplanetary disc structure, chemistry and late-stage infall

JWST Co-Investigator on 6 awarded programmes spanning Cycles 1–4 (PIs Bik, Kuhn, Guarcello ×2, Benisty and Ramirez-Tannus), studying externally irradiated and starburst environments

Outreach

Mar 2026 ‘Think Space Lecture – ‘[Unveiling our origins](#)’

Mar 2016/17/18 Cambridge Science Festival, Institute of Astronomy – public talks on planet formation and a computer game teaching gravity ([Gravity Hero](#))